

"van Son et al. 2022a/b" is not a unique identifier: a citation-resolution hazard in the COMPAS literature

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Finding. Each of the labels "van Son et al. 2022a" and "van Son et al. 2022b" refers to **two different papers**, depending on which paper is doing the citing. Three papers by the same first author appeared within about a year, with overlapping author lists and one preprint-to-journal transition among them; each citing paper assigned the a/b suffixes in its own bibliography, and the assignments disagree. Every paper is internally consistent — but the string is not globally unique, so resolving a deferred parameter by author-year-letter lands on the wrong source.

The three papers (third author is a clean discriminator)

Journal ref	Short title	arXiv	DOI suffix	3rd author
ApJ 931, 17	The redshift evolution of the BBH merger rate: a weighty matter	2110.01634	ac64a3	Callister
ApJ 940, 184	No Peaks without Valleys	2209.13609	ac9b0a	Renzo
ApJ 948, 105	The locations of features ... robust against uncertainties in the metallicity-dependent cosmic SFH	2209.03385	acbf51	Chruślińska

What each paper calls each paper — read from their own reference lists

Paper being cited	ApJ 940,184 calls it	ApJ 948,105 calls it	van Son 2024 calls it
ApJ 931, 17	2022b	2022a	2022b
ApJ 940, 184	(itself)	2022b	2022a
ApJ 948, 105	2022a	(itself)	2023

ApJ 948, 105 carries a **third** label: cited as "2022a" while it was arXiv:2209.03385, then as "**2023**" once it appeared in ApJ. Same paper, three names.

Three distinct failure modes

- 1. Circular.** ApJ 940, 184 states: "*We refer the reader to the methods section of van Son et al. (2022b) for a more detailed description of the remainder of the adopted physics parameters.*" Resolved by label, that points back at ApJ 940, 184 itself. By its reference list it means **ApJ 931, 17**.
- 2. Transitive — the near source is the wrong one.** ApJ 948, 105 uses "*the simulations behind Figure 1 from van Son et al. (2022b)*" = ApJ 940, 184. But ApJ 940, 184 says Figure 1 was produced "*using COMPAS version v02.26.03 with the exact same settings as the fiducial model for isolated binary formation from van Son et al. (2022b)*" — and in its numbering that is **ApJ 931, 17**. The physics therefore traces two papers away, with the letter meaning a different thing at each hop.
- 3. Right paper, wrong model.** Even landing correctly on ApJ 940, 184, its *reference model* is not its Figure 1 model: the reference model fixes $\beta_{\text{acc}} = 0.5$ and $\zeta_{\text{eff}} = 6.0$, whereas Figure 1 uses thermal-timescale-limited accretion and $\zeta_{\text{eff}} = 6.5$. Inheriting the reference-model values would import the wrong accretion treatment and the wrong stability threshold.

Two independent checks confirming the resolution

The reading does not rest on the reference lists alone. Two unusual values are attributed by one paper and confirmed verbatim in the other:

ApJ 940, 184: "*We adopt $\zeta_{\text{eff}} = 6.0$... compared to $\zeta_{\text{eff}} = 6.5$ in van Son et al. (2022b).*" — ApJ 931, 17, in its own words: "*We assume $\zeta_{\text{ad}} = 2$ for main sequence donors, $\zeta_{\text{ad}} = 6.5$ for Hertzsprung gap donor stars (Ge et al. 2015).*"

ApJ 940, 184 on its Figure 1 model: "*C = 10 is a constant factor...*" — ApJ 931, 17: "*the accretion rate is limited to ten times the thermal rate of the accretor.*"

Why this is a result, not a nuisance

Any extraction pipeline — LLM-assisted or otherwise — that resolves deferred parameters by author-year-letter will mis-attribute physics here, and will do so **silently**, emitting a plausible value from the wrong paper. The three modes are structurally different (circular, transitive, intra-paper), so none is caught by a single guard. Resolution must go through the citing paper's own reference list, or better a DOI or arXiv ID.

Rule now applied in the ledger. A value may be inherited from another paper only when (a) the citing paper explicitly directs the reader there and ties the pointer to that parameter or to the source's methods section; (b) the source is in hand, so its own sentence can be quoted; (c) the citing paper does not list that parameter among its own stated changes; and (d) no version gap or documented code correction separates them on that row. Inherited cells are rendered distinctly and carry **both** sentences — the deferral and the source's own words — each verified against its own paper's text. **Recommendation:** label these rows by journal reference or arXiv ID, never by the a/b letter.

Every quotation and reference-list line above was extracted from the published PDFs with pdftotext and checked programmatically for verbatim presence; the ledger build fails if any quote drifts from its source (1,554 currently pass). Not verified: how AAS or ADS mechanically assigns the a/b suffix — the divergence is documented here, its cause inferred.